



THE UNIVERSITY
OF BRITISH COLUMBIA

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Dear Search Committee,

Thank you for considering my application for the tenure track position in the Biological Sciences Division at University of Chicago. I am an evolutionary geneticist whose goal is to understand the nature and tempo of adaptation to changing environments, from genotype to phenotype. Conceptually, I am drawn to the vast diversity of plants and the opportunities they present for characterizing factors that limit the rate of adaptation, while applications of this work are crucial for informing global strategies of efficient food production and human health. I use population genomic approaches bolstered by experimental and theoretical inference to address these issues, and look forward to contributing my perspective to the diverse community at UChicago.

My research on plant evolution has spanned timescales, taxa, and approaches. As an undergraduate researcher with Dr. Brian Husband at the University of Guelph, I combined extensive field collections, flow cytometry, and phylogenetic analyses to study the evolutionary dynamics of unreduced gamete production—work that produced two influential publications for understanding drivers of polyploid formation. In my Ph.D. at the University of Toronto with Drs. Stephen Wright and John Stinchcombe, I established a novel research program on the population genomics of herbicide resistance. I led international, interdisciplinary collaborations between Molecular Biologists and Weed Scientists, in which we developed genomic resources for studying herbicide resistance in the noxious agricultural weed, *Amaranthus tuberculatus*. Out of this effort, I have conceived and led five papers in *PNAS*, *Annu. Rev. Plant Biol.*, *Mol. Ecol.*, *Evolution*, and *eLife*. As a postdoctoral researcher supported by two independently funded fellowships at the University of British Columbia, I sought out collaborations with Drs. Sarah Otto and Loren Rieseberg to expand my research program to study the temporal dynamics of contemporary adaptation—developing theoretical models and statistical approaches for powerful inference of an unprecedented dataset spanning 200 years of environmental change (accepted at *Science*). My work has made strides in our understanding of the link between genomic change and our changing environment. I see the University of Chicago, with its emphasis on collaborative and innovative approaches to foundational and translational research, as the ideal place to grow my research program on plant adaptation to contemporary environments.

Having been greatly inspired by both my mentors and mentees, I am highly motivated to provide students and postdocs with broad training to accomplish their goals in academia, industry, and beyond, while fostering a diverse and inclusive community. As a leader in my Ph.D. department, I advocated for student mental health and well-being during my time as student president, obtaining increased student stipends to combat escalating costs of living. During my time at UBC, I have been working to reinforce a community that supports and retains diverse talent, by serving as a member of the Botany EDI committee, participating in the Biology Undergraduate Diversity in Research Program, and facilitating first time research experiences as a work-study mentor. As a curriculum developer and instructor for a graduate student Bioinformatics course, I implement active and engaged learning strategies to provide an accessible and approachable introduction to programming, a foundational skill many students find daunting.

At University of Chicago, I would be keen to continue to engage in diverse collaborations within and across departments, from the Department of Ecology & Evolutionary Biology to Molecular Biology & Genetics and Public Health Sciences, while continuing to establish local and international ties across institutes. I look forward to contributing my teaching and mentorship experience, inclusive pedagogical philosophy, and plant

evolutionary genomic expertise to strengthen the community in the Biological Sciences Division. Please find enclosed my full application that demonstrates the breadth of my research program along with my list of reference contacts below.

Sincerely,
Julia M. Kreiner



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